

PROBABILITY

Date: 17/10/24

Stats $\rightarrow$ science of collecting & analyzing.

Data $\rightarrow$ collection of facts & figures

Quantitative Data / Qualitative Data.

Discrete Data $\rightarrow$ specific value.

Continuous Data $\rightarrow$ any value within range. (temp)

Unstructure ko Structure krna.

RAW $\rightarrow$ Organized (Table form)

★ Slide Example 1 ★

~~26-45~~ 27 50

Class Interval	Tally Marks	Frequency
25-29 26-30		7
30-35		12
36-40		11
41-45		12
46-50		10
51-55		6
56-60		2
61-65		4
66-70		1
71-75		
Total		65

Ex 2 $\rightarrow$ 1.6 $\rightarrow$ 2

Date: _____

1) Find Maximum & Min Num. .

2) Range = Max - Min

$$\text{Width } ~~\text{Class}~~ ~~\text{Interval}~~ = \frac{\text{Range}}{\text{No of Classes}}$$

21/10/24

We can choose between 5 - 15 (Classes)

7 Class (Ex 3)

1 class ki upper limit doosre class ki lower limit same $\rightarrow$ Continuous.

1 decimal $\rightarrow$ lowest value = 0.1 $\Rightarrow$ 0.1 $\Rightarrow$ 0.05

2
Class Boundary

Example 2 $\rightarrow$

Class Interval	Tally Marks	Frequency
1.6 - 2		2
2.1 - 2.5		2
2.6 - 3		5
3.1 - 3.5		5
3.6 - 4		5
4.1 - 4.5		4
4.6 - 5		2
		<u>2</u>
		<u>40</u>

Date: _____

Example 3 →

(7) Class

Class Interval	Tally	F	Class Boundaries	Mid Value
85-91	I	1	84.5 - 91.5	88
92-98	III	3	91.5 - 98.5	95
99-105	IIII	5	98.5 - 105.5	102
106-112	III	3	105.5 - 112.5	109
113-119	IIIIII	12	112.5 - 119.5	116
120-126	IIIIIIII	13	119.5 - 126.5	123
127-133	IIII	5	126.5 - 133.5	130
134-140	IIII	5	133.5 - 140.5	137
141-147	III	3	140.5 - 147.5	144

23/10/24

Data Analytics.

1) Descriptive Analytics → Process of Analyzing historical data
 ⊗ Summarization, Data Visualization, Historical Analysis, Reporting.

Month	Year 1 Sales	Year 2 Sales	Year 3 Sales
January	\$ 5000	5200	5400
Feb	5100	5300	5500
March	5050	5150	5350
April	5500	5550	5700
May	5600	5700	5850
June	5650	5800	6000
July	5950	6000	6050

August	5800	5850	6000
September	6000	6200	6250
October	6050	6250	6450
November	6150	6400	6600
December	9000	9550	10000

24/10/24

Q) Find the aggregate by data categorise wise

Month	Clothing Sales	Footware Sales	Accessories Sales	Total Sale
January	3500	1200	300	5000
February	3600	1300	300	5200
March	3550	1250	300	5100
	10,650	3750	900	

Calculate Growth Rate From January to February

$$\frac{\text{Total Feb Sale} - \text{Total Jan Sale}}{\text{Jan Total Sale}} \times 100$$

$$\Rightarrow \frac{5200 - 5000}{5000} \times 100 \Rightarrow 4\%$$

Calculate Growth Rate from Febu to March

~~5100~~
$$\frac{5100 - 5200}{5000} = -1.92$$

Summarize by Percentage Contribution

Cloth Jan	Cloth Feb	Cloth March
$\frac{3500}{5000} \times 100 = 70\%$	$\frac{3600}{5000} \times 100 = 72$	$\frac{3550}{5100} \times 100 = 69.61$

Next Thursday Quiz

34

Amperical Important

Date: _____

$$\text{Footwear Jan } \left\{ \frac{1200}{5000} \times 100 = 24\% \right.$$

$$\text{Footwear Feb } \left\{ \frac{1300}{5200} \times 100 = 25\% \right.$$

$$\text{Footwear March } \left\{ \frac{1250}{5100} \times 100 = 24.51 \right.$$

$$\text{Accessories Jan } \left\{ \frac{300}{5000} \times 100 = 6\% \right.$$

$$\text{Accessories Feb } \left\{ \frac{300}{5200} \times 100 = 5.77 \right.$$

$$\text{Accessories March } \left\{ \frac{300}{5100} \times 100 = 5.88 \right.$$

Diagnostic Analytics and Predictive Analytics

30/10/24

Work in CN $\rightarrow$ Look at that Important.

Arithmetic Mean, Median Mode

31/10/24

Jahaan Xtreme Value, Wahaan Median use kien gay.
25-30 30-35 $\rightarrow$ Continuous.
Discrete k liye class boundary.

Mode $\rightarrow$ Most Repeated Values in Data

Measure of Variability $\rightarrow$

Range, Quartile Deviation, Mean Deviation, Variance, Standard Deviation

MD

$$\text{Group } \rightarrow \frac{\sum (x - \bar{x})}{n} \quad \left\{ \begin{array}{l} \bar{x} \rightarrow \text{Sum} \\ \end{array} \right.$$

Date: 4/11/24

Mean Deviation about Mean Median
 $2, 3, 4, 2, 4$ $MD(\bar{x}) = \frac{\sum |x - \bar{x}|}{n}$

$$\bar{x} = \frac{\sum x}{n} = \frac{15}{5} = 3$$

x	$x - \bar{x}$	$ x - \bar{x} $
2	$2 - 3 = -1$	1
3	$3 - 3 = 0$	0
4	$4 - 3 = 1$	1
2	$2 - 3 = -1$	1
4	$4 - 3 = 1$	1
		<u>4</u>

$$\Rightarrow \frac{4}{5} = 0.8$$

Mean Dev Median
 $MD(\tilde{x}) = \frac{\sum |x - \tilde{x}|}{n}$

1) Ascending Order

2, 2, 3, 4, 4.

$$\frac{n+1}{2} = \frac{6}{2} = 3$$

Median = 3

x	$ x - \tilde{x} $	$ x - \tilde{x} $
2	$2 - 3 = -1$	1
2	$2 - 3 = -1$	1
3	$3 - 3 = 0$	0
4	$4 - 3 = 1$	1
4	$4 - 3 = 1$	1
		<u>4</u>

$$\Rightarrow \frac{4}{5} = 0.8$$

mean Deviation about median

$$\frac{\sum |x - \tilde{x}|}{n}$$

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7, 15, 13, 89, 200
 → 7, 13, 15, 89, 200

$$\frac{n+1}{2} = \frac{6}{2} = 3$$

Median

[15] → Arrange kare k baad wali value ko minus karen gay

X	$x - \tilde{x}$	
7	$7 - 15 = -8$	8
13	$13 - 15 = -2$	2
15	$15 - 15 = 0$	0
89	$89 - 15 = 74$	74
200	$200 - 15 = 185$	185
		<u>269</u>

$$\Rightarrow \frac{269}{5} = 53.8$$

3rd Place mein 15 value se minus.

Mean Dev about Mean

Class	f	$f x$	$x - \bar{x}$	$ x - \bar{x} $	
2	4	8	$2 - 4.5 = -2.5$	2.5	1.4
3	5	15	$3 - 4.5 = -1.5$	1.5	0.4
4	6	24	$4 - 4.5 = -0.5$	0.5	0.6
5	3	15	$5 - 4.5 = 0.5$	0.5	1.6
	<u>18</u>	<u>62</u>			

$$\bar{x} = \frac{\sum f x}{\sum f} = \frac{62}{18} = 3.4$$

Mean -

Class - Mean

≤

$$\frac{\sum f |x - \bar{x}|}{\sum f} = 5.6$$

6 / 11 / 24

2, 3, 4, 5, 8

$$M_1) (\bar{x}) = \frac{\sum |x - \bar{x}|}{n}$$

$$M_1) (\tilde{x}) = \frac{\sum |x - \tilde{x}|}{n}$$

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Class	f
2	5
3	6
4	7
5	8

$$M.D.(\bar{x}) = \frac{\sum f |x - \bar{x}|}{\sum f}$$

$$M.D.(\tilde{x}) = \frac{\sum f |x - \tilde{x}|}{\sum f}$$

CI	f
2-3	5
4-5	6
6-7	3
8-9	4

$$M.D.(\bar{x}) = \frac{\sum f |x - \bar{x}|}{\sum f}$$

$$M.D.(\tilde{x}) = \frac{\sum f |x - \tilde{x}|}{\sum f}$$

Standard Deviation.

σ^2 = Variance

σ = Standard Deviation

$$\sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$$

$$\bar{x} = \frac{\sum x}{n} = \frac{22}{5} = 4.5 \Rightarrow \text{Matlab has value for mean se minus.}$$

x	$x - \bar{x}$	$ x - \bar{x} $	
2	2 - 4.5	2.5	6.25
3	3 - 4.5	1.5	2.25
4	4 - 4.5	0.5	0.25
5	5 - 4.5	0.5	0.25
8	8 - 4.5	3.5	12.25
			21.25

$$\sigma^2 \Rightarrow \frac{21.25}{5} \Rightarrow 4.25$$

$$\sigma = \sqrt{4.25} \Rightarrow 2.06$$

Ques $\rightarrow$ Consistency / Which better

Grouped Data

Date: _____

$$\frac{\sum f(x-\bar{x})^2}{\sum f}$$

Class Interval mean Mid Point (x) hoga

CI	f	Mid	fx	$ x-\bar{x} ^2$	
2-3	5	2.5	12.5	10 7.07	$\frac{\sum fx}{\sum f} = \frac{93}{18} \Rightarrow \boxed{5.16}$
4-5	6	4.5	27	50.625 0.43	
6-7	3	6.5	19.5	14.1 1.82	
8-9	4	8.5	34	65.025 11.15	
				142.55	

$$\begin{array}{l} f(x-\bar{x})^2 \\ 7.07 \\ 0.43 \\ 1.82 \\ 11.15 \end{array} \quad \left. \begin{array}{l} f(x-\bar{x})^2 \end{array} \right\}$$

11/11/24

5, 15, 16, 20, 24, 25, 26, 27, 30, 30, 31, 32, 32
34, 35, 38, 41, 43, 66

1) Arrange in Ascending Order.

a) 5 Num Sum

1) Minimum = 5

2) Maximum = 66

3) First Quartile = 23 Q1

4) Second Quartile = 30.5 Q2

5) Third Quartile = 36.5 Q3

- 5 Number Summary
- Inner Quartile ✓
- Range
- Quartile Deviation
- Lower Limit, Upper Limit
- Outlier
- Boxplot

Second Quartile $\Rightarrow \frac{n}{2}$ and $\frac{n+1}{2} \Rightarrow \frac{20}{2}$ and $\frac{22}{2}$

10th and 11 value.

30 and 31

$$\Rightarrow \frac{30+31}{2} \Rightarrow 30.5$$

$\frac{n+1}{2}$ and $\frac{n}{2}$

~~$\frac{10}{2}$~~ ~~$\frac{12}{2}$~~

5th & 6th value

$$\frac{21+25}{2}$$

$\Rightarrow 23$ first Quar

$\frac{n+1}{2}$ and $\frac{n}{2}$

$\frac{10}{2}$ and $\frac{12}{2}$

5th & 6th Value

$$\frac{35+38}{2}$$

36.5

Inner Quartile Range

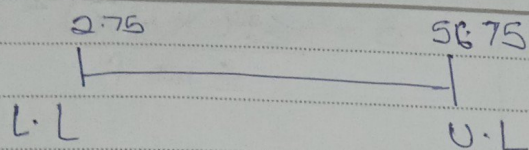
$$Q_3 - Q_1 = 36.5 - 23 = 13.5$$

Quartile Deviation $\frac{Q_3 - Q_1}{2} \Rightarrow \frac{13.5}{2} \Rightarrow 6.75$

Lower Limit $Q_1 - \frac{1.5}{2} IQR \Rightarrow 23 - 1.5(13.5)$
 $\Rightarrow 2.75$

Upper Limit $Q_3 + \frac{1.5}{2} IQR \Rightarrow 36.5 + 1.5(13.5)$
 $\Rightarrow 56.75$

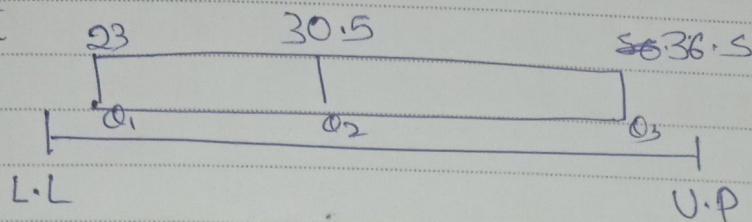
Outlier



Date: _____

66 is outlier

Box Plot



13/11/24

Coefficient of Variation

$$\frac{\sigma}{\bar{x}} \times 100$$

$\sigma^2 = \text{variance}$

$\sigma = \text{Standard Deviation}$

Q

Coefficient of Variation

Compare (Different Unit)

Bat 1	
Bowl	

$$\frac{\sigma}{\bar{x}} \times 100$$

Standard Deviation (Previous Page)
Same Unit

Bat 1	
Bat 2	

1) 5, 7, 8, 9

Class	f
2	5
3	6
4	7

$\sigma^2 = \text{variance}$
 $\sqrt{\sigma^2} = \text{variance}$
 $\sigma = \text{St. Devia}$

Date: _____

$$\sigma (\text{ungrouped}) = \frac{\sum (x - \bar{x})}{n}$$

$$\sigma = \frac{\sum f(x - \bar{x})^2}{\sum f}$$

3)

CI	f
2-3	5
4-5	8
6-7	9

Variance = 6

Mean = 4

Find Coefficient of Var

$$\frac{\sqrt{6}}{4} \times 100$$

$$C.V = 0.38$$

$$\sigma = 4$$

Mean

$$\frac{\sigma}{\bar{x}} \times 100$$

Data Given Make Frequency Distribution
 Measure of Center / Tendency
 • Mean, Median, Mode

18/11/24

Disparity when dataset is unbalance.
 Symmetrical $\rightarrow$ Mean = Median = Mode

Positive Skew

$$\text{Mean} > \text{Median} > \text{Mode}$$

Negative Skew

$$\text{Mode} > \text{Median} > \text{Mean}$$

Pearson 2nd Coeffic
$$\frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

Find Out Co-efficient of Skew.

34, 35, 40, 50, 190, 200.

$$\text{Mean} = \frac{549}{6} \Rightarrow 91.5$$

$$\text{Median} = \frac{6}{2} = 3^{\text{th}} \text{ and } \frac{6}{2} = 4^{\text{th}}$$

$$\frac{40 + 50}{2} \Rightarrow 45.$$

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$= \sqrt{\frac{32347.5}{6}} = \sqrt{5391.25}$$

$$\sigma \Rightarrow 73.4$$

$$\frac{3(91.5 - 45)}{73.4}$$

C. Skew $\Rightarrow 1.9$, it shows data is extremely skew.

Date:

Near Symmetrical $\rightarrow -0.5$ to 0.5
 Negative Skew $\rightarrow -1$ to -0.5
 Positive Skew $\rightarrow 0.5$ to 1
 Lower than -1 or greater than 1 (+ve) data are extremely skew

X	(x - $\bar{x}$)	(x - $\bar{x}$) ²
34		
35		
40		
50		

X	(x - $\bar{x}$)	(x - $\bar{x}$) ²
34	-57.5	3306.25
35	-56.50	3192.25
40	-51.5	2652.25
50	-41.5	1722.25
190	98.5	9702.25
200	108.5	11772.25

3306.25
 3192.25
 2652.25
 1722.25
 9702.25
 11772.25
 32347.5

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70, 72, 75, 78, 80, 82, 85, 87, 90,
93, 150

20/11/24

KURTOSIS

Mesity Lepto, Meso, Platy kurtic

Steps-

- 1) Calculate Mean
- 2) Standard Deviation
- 3) Fourth Moment
- 4) Calculate kurtosis

kurtosis > 3 Lepto
 kurtosis $= 3$ Meso
 kurtosis < 3 Platy

21/11/24

$$\text{Kurtosis} = \frac{\text{Fourth Moment}}{s^4}$$

$$\text{Fourth Moment} = \frac{\sum (x - \bar{x})^4}{n}$$

X = 3, 5, 7, 8, 9, 10, 12, 15

Mean = 8.625

G.D		$x - \bar{x}$	$(x - \bar{x})^2$	$(x - \bar{x})^4$
	3	-5.625	31.64	1001.0896
	5	-3.625	13.14	172.6596
	7	-1.625 -1.625	2.64	6.9696
	8	-0.625	0.39	0.1521
	9	0.375	0.14	0.0496
	10	1.375	1.89	3.5721
	12	3.375	11.39	129.7321
	15	6.375	40.64	1651.6096
			<u>101.87</u>	<u>2965.8043</u>

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$$\text{Var} = \frac{\sum (x - \bar{x})^2}{n} \Rightarrow \frac{101.87}{8} \Rightarrow 12.73$$

$$\text{Var}^2 = F_{s^4} = 162.05$$

$$\text{Fourth Moment} = 370.7255 \leftarrow \frac{2965.80}{8}$$

$$\text{kurtosis} = \frac{370.7255}{162.05}$$

$$\boxed{2.29}$$

That The given data is Platy kurtic since kurtosis < 3

25/11/24

Bar vs Line Graph (Index)

Numerical + Theoretical.

Bar $\rightarrow$ Used to Compare Categorical or Nominal Data where categories are distinct, represented in rectangular bar.

Line Graph $\rightarrow$ Represent continuous data, points are plotted then connected

Dual Axis Chart

Pie Chart represent proportional data used to show the percentage of particular data from the whole pie

Relative Frequency.

$$\left. \begin{array}{l} \text{Relative Frequency} \\ \text{Pie Chart} \end{array} \right\} \frac{\text{Given Data}}{\text{Total Data}} \times \cancel{100} 360$$

* Box Plot, 5 Num

Date: _____

Pie Chart

27/11/24

Histogram (Skewness)
Equal Class Interval

Hist 64 → 1551

Minimum Value → Standard

Histogram → Continuous → Class Boundaries
Doughnut Charts

$$\frac{n}{\text{Total}} \times 100$$

2/12/24

Counting 1) Multiplication

$$n_1 \times n_2$$

Coin and Dice

$$2 \times 6 = 12 \text{ ways (Outcome)}$$

0 1 2 3 4 5 6
4 digit even number, no repetition

$$\frac{6 \times 5 \times 4 \times 1}{0} = 120$$

left side per 0 ko
count nhi krni gey.

$$\frac{5 \times 5 \times 4 \times 1}{1} = 100$$

yahaan 0
nhi ayege

↳ yahaan 2 ayege.

$$\frac{5 \times 5 \times 4 \times 1}{4} = 100$$

$$\frac{100}{6} = \frac{100}{420}$$

$$0! = 1$$

Date: _____

Permutation (Arrangement) (how many ways)
 abc events $\rightarrow$ abc, acb, bac, bca, cab, cba
 Row. $n! \rightarrow 3 \times 2 \times 1 = 6$ combinations.

In Circle $(n-1)!$
 $(3-1)! = 2$

$n \rightarrow$
 $r \rightarrow$ Object

nPr

Repetition

$$\frac{n!}{n_1! n_2! n_3!}$$

w k ander se.

Combination (Selection) nCr

$$\text{Probability} = \frac{\text{Interest}}{\text{Outcome}} = \frac{\text{Favourable}}{\text{Sample Space}} \quad 0 \leq P_r \leq 1$$

Two Coin tossed, Probab atleast 1 head.
 $S = \{HH, HT, TH, TT\}$
 $\frac{3}{4} \Rightarrow 0.75$

Probability
 mein 0 ✓
 se 1 k
 darman
 answer ayege

E/Amparimash

Date: _____

Q1) A dice is loaded in such a way that each odd number is twice as likely to occur to each even number. Find $P(G)$, where G is number greater than occur on a single roll of the dice.

Dice = $\{1, 2, 3, 4, 5, 6\}$

Sample = $\{1, 1, 2, 3, 3, 4, 5, 5, 6\}$ 9.

Greater than 3 $\rightarrow$ 4, 5, 5, 6

$\frac{4}{9}$ Ans.

Q2) In a poker hand, consisting of 5 cards. Find the probability of holding 2 aces & 3 jacks.

Total Cards = 52

Sample Space = $52C5 = 2598960$

$4C2 \times 4C3 = 24$.

$\frac{24}{2598960}$

52 total
13 $\rightarrow$ Diamond R
13 Spade B
13 Club B
13 Heart R

(D, S, C, H) Ace
(D, S, C, H) king Diag
(D, S, C, H) Queen Diag
(D, S, C, H) Jack Diag
(9) 10-2

Additive Rule

Date: _____

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

What is the probability of getting a total 7 or 11 when a pair of dice is tossed.

(1,1) (1,2) 1,3 1,4 1,5 1,6
 2,1 2,2 2,3 2,4 2,5 2,6
 3,1 3,2 3,3 3,4 3,5 3,6
 4,1 4,2 4,3 4,4 4,5 4,6
 5,1 5,2 5,3 5,4 5,5 5,6
 6,1 6,2 6,3 6,4 6,5 6,6

For 7 $\rightarrow \frac{6}{36}$

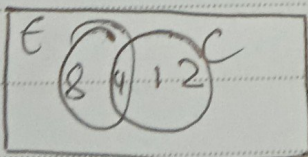
For 11 $\frac{2}{36}$

$$\frac{6}{36} + \frac{2}{36} - 0 = \frac{8}{36}$$

What is probability of atleast 8

$$\frac{15}{36}$$

(2.49 2.71)



$$\begin{aligned}
 P(E \cup C) &= \frac{4}{24} + \frac{16}{24} - \frac{4}{24} \\
 &= \frac{24}{24} = 1
 \end{aligned}$$

$$P(A) + P(A') = 1$$

If the probabilities that an automobile mechanic will 3, 4, 5, 6, 7 or 8 or more cars on any given work day are respectively 0.12, 0.19, 0.28, 0.24, 0.10, and 0.07

What is probability that he will service at least 5 cars on his next day at work.

Humien 8 k uper ki probability nhi pata toh 3 or 4 ki probab se Total minus.

$$1 - P(3 \text{ or } 4) = 0.31 \quad (0.12 + 0.19)$$

$$1 - 0.31 = \boxed{0.69}$$

2, 2, 2, 3, 4, 5, 9

D5 $\rightarrow$ Decile

$$5 \times \frac{(n+1) \text{ (odd)}}{10}$$

$$5 \times \frac{(7+1)^{\text{th}}}{10} \Rightarrow \frac{40}{10} = 4^{\text{th}} \text{ value}$$

$$P_{66} = \frac{66}{100}$$

Answer $\rightarrow 3$

Date: _____

$$1 + \frac{1}{f} \left(\frac{20}{5} - 0.1 \right)$$

$$\text{Dec} \rightarrow 1 + \frac{1}{f} \left(7 \frac{20}{10} - 0.1 \right)$$